



PHILOMATH NETWORKS

**PHILOMATH
NETWORKS PVT. LTD**



6-Days Hands-On Workshop on "Big Data Certification Training Course"

**Get ready to earn your Big Data
Certification with us!**

Redefining Learning
Beyond Boundaries



**Phone Number
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About Philomath Networks



At Philomath Networks Pvt. Ltd., we specialize in organizing transformative events for schools, organizations, and communities, fostering a culture of exploration, innovation, and growth. From expert-led workshops to competitive events, we create immersive learning experiences that inspire and empower individuals to go beyond boundaries. Partner with us to bring impactful learning to your community!



Partner With Us

Collaborate with Philomath Networks to bring innovation, engagement, and excellence to your institution or organization. Let's work together to create an ecosystem of lifelong learning and impactful experiences!

What Sets Us Apart?

Comprehensive Learning Approach – We blend academic, non-academic, and skill-based experiences through workshops, competitions, expert-led sessions, and immersive programs tailored to real-world applications.

End-to-End Event Management – From conceptualization to execution, we ensure seamless organization of competitions, tech fairs, hackathons, leadership programs, and customized training sessions that align with institutional goals.

Industry & Academic Bridging – We foster collaborations between institutions, corporates, and experts, ensuring that participants gain practical exposure, networking opportunities, and insights into emerging trends beyond traditional education.

Customizable & Scalable Programs – Whether it's a school-level science fest, university hackathon, corporate leadership training, or specialized certification courses, we tailor our events to meet the unique needs of every institution.

Proven Impact & Engagement – Our interactive approach ensures higher participation, deeper learning retention, and enhanced skill development, making every event a transformative experience.

Course Overview



▶ Course Objective

This intensive training program provides a comprehensive understanding of Big Data concepts, tools, and technologies. Participants will explore data processing, distributed computing, and analytics using Hadoop, Spark, and real-time data processing frameworks. Through hands-on labs and real-world case studies, learners will gain practical experience in handling large-scale data, optimizing workflows, and deriving insights. By the end of the course, participants will be proficient in managing and analyzing Big Data, preparing them for roles in data engineering and analytics.

▶ Course Outcome

✓ Big Data Fundamentals – significance, and applications.

✓ Analytics & Visualization – Real-time streaming

✓ Data Storage & Management – Hadoop, HDFS, and distributed storage

✓ Security & Governance – Data protection and compliance .

✓ Big Data Processing – Spark, MapReduce, and data pipelines.

✓ Hands-on Projects – Practical case studies

Training Modules

The training modules provide a hands-on learning experience in core Big Data concepts, distributed computing, and data processing. Participants will begin with Big Data fundamentals, storage frameworks, and processing techniques using platforms like Hadoop, Spark, and cloud-based solutions. They will then explore real-time analytics, data security, and workflow automation to enable scalable and efficient data solutions. Each module is structured around real-world scenarios, ensuring practical skill development. Additionally, learners will gain insights into best practices for optimizing data processing, managing large datasets, and enhancing overall analytics efficiency.

Module 1: Introduction to Big Data & Ecosystem

- Understanding Big Data and its importance
- Overview of Big Data technologies and tools
- Hadoop ecosystem and architecture
- Distributed computing and storage fundamentals
- Introduction to batch and real-time processing
- Use cases and applications of Big Data



Module 2: Data Storage & Processing with Hadoop

- HDFS architecture and file system commands
- MapReduce programming model and execution flow
- YARN resource management and job scheduling
- Working with Hadoop ecosystems: Hive, Pig, and HBase
- Data ingestion using Sqoop and Flume
- Optimizing Hadoop performance and scalability



Module 3: Data Processing with Apache Spark

- Introduction to Apache Spark and its advantages
- Spark architecture, RDDs, and DataFrames
- Spark SQL for structured data processing
- Implementing real-time streaming with Spark Streaming
- Spark MLlib for machine learning in Big Data
- Performance tuning and optimization in Spark



Training Modules

As the course progresses, participants will explore data pipeline automation with Apache NiFi, monitoring with Apache Ambari and Prometheus, and security best practices in Big Data environments. The final modules focus on data governance, cost optimization, and compliance, ensuring learners are industry-ready to implement scalable, secure, and efficient data solutions. By the end of the training, participants will have the confidence to design, deploy, and manage Big Data architectures in real-world environments. This comprehensive approach will prepare them for Big Data roles in leading tech industries.

Module 4: Real-Time Data Processing & Analytics

- Introduction to real-time data pipelines
- Apache Kafka for distributed messaging
- Apache Flink and Storm for real-time analytics
- Integration of Spark Streaming with Kafka
- Data visualization techniques for Big Data
- Case studies on real-time analytics

Module 5: Big Data in Cloud & Data Engineering

- Cloud-based Big Data solutions (AWS, Azure, GCP)
- Serverless computing for Big Data workloads
- Managing data lakes and warehouses in the cloud
- ETL pipelines and data transformation
- Best practices for scalable and cost-effective solutions
- Security and compliance in cloud-based Big Data

Module 6: Capstone Project & Certification

- End-to-end Big Data project development
- Data pipeline design and implementation
- Processing, analyzing, and visualizing large datasets
- Performance tuning and optimization strategies
- Best practices for Big Data applications
- Certification assessment and career guidance



Training Schedule & Format



The training schedule is structured as a 6-day intensive Big Data workshop, providing a hands-on learning experience. Each day focuses on a key Big Data module, starting with foundational concepts and progressing to advanced topics like data storage, processing, security, and real-time analytics. The workshop includes interactive sessions, hands-on labs, and real-world data problem-solving to reinforce learning. Participants will work on mini-projects and practical assignments daily, ensuring a strong grasp of Big Data tools and best practices. The final day is dedicated to a capstone project, performance optimization, and a certification assessment, equipping learners with industry-ready Big Data skills.

Day	Module	Focus Area	Duration
Day 1	Introduction to Big Data & Ecosystem	Concepts, Tools, Use Cases	5hrs
Day 2	Data Storage & Processing with Hadoop	HDFS, MapReduce, Hive	5hrs
Day 3	Data Processing with Apache Spark	RDDs, DataFrames, Streaming	5hrs
Day 4	Real-Time Data Processing & Analytics	Kafka, Flink, Visualization	5hrs
Day 5	Big Data in Cloud & Data Engineering	Cloud, ETL, Security	5hrs
Day 6	Capstone Project & Certification	Project, Optimization, Exam	5hrs

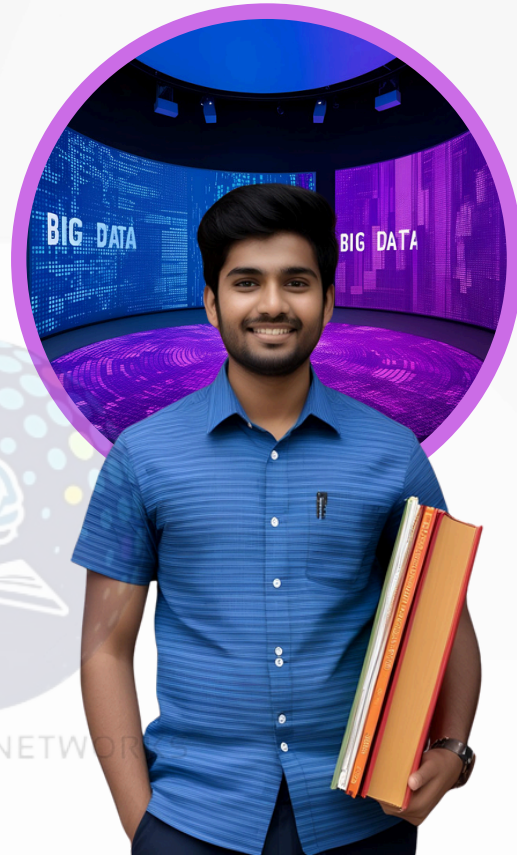
Through hands-on exercises and case studies, participants will develop essential Big Data skills, explore best practices, and tackle real-world data processing challenges. The workshop includes live demonstrations, guided projects, and interactive labs to enhance learning. A collaborative format encourages discussions, troubleshooting, and peer learning. By the end, attendees will have a strong Big Data foundation, enabling them to process large datasets, optimize data workflows, and manage scalable data infrastructure effectively.

How to Get Involved



Steps to Enroll in Training Programs

- Contact us via email or phone.
- Share your preferred course, schedule, and any specific requirements.
- Our team will review your request and confirm the booking.
- You will receive confirmation details via email or call .



Contact Information

For any questions about the training program, please reach out to:

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